

Sustainable Petroleum Refining: Biodiesel Production

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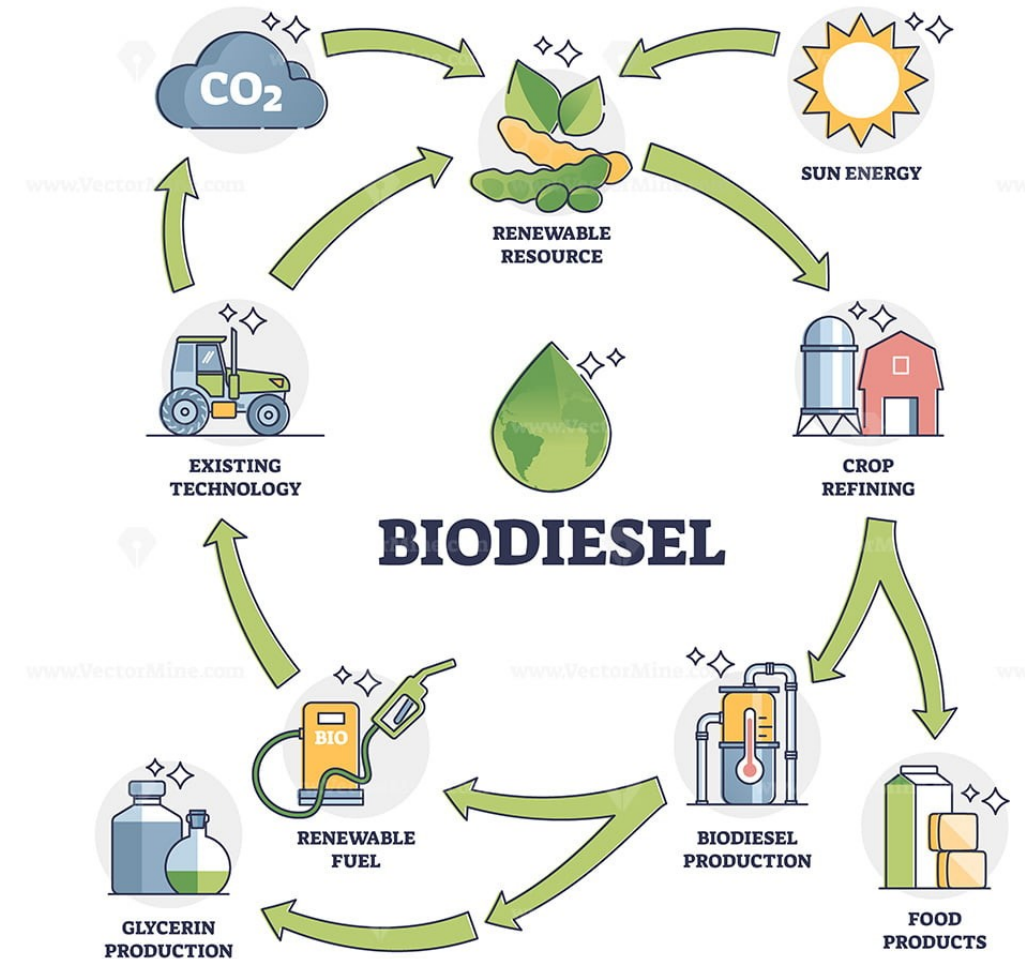
Problem Definition & Significance

- Heavy global reliance on petroleum diesel for transport, agriculture, and industry
- Diesel is nonrenewable; demand & supply pressures rising (IEA, 2023)
- Major contributor to greenhouse gases (climate change) & air pollution → health impacts



Why Biodiesel

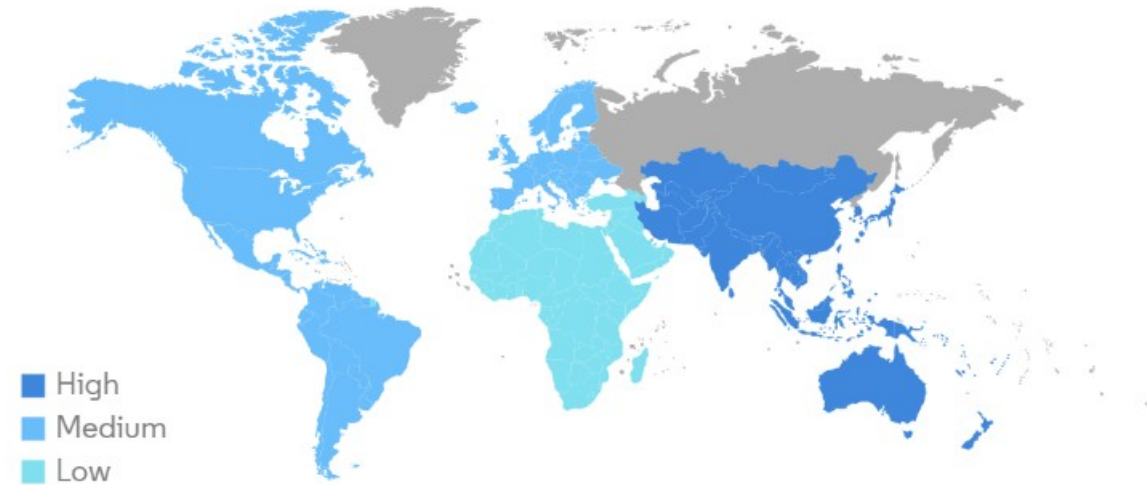
- Biodiesel = renewable, biodegradable, up to 78% lower CO₂ vs. petroleum diesel (DOE, 2024)
- Derived from plant oils, animal fats, or waste cooking oils (WCO)
- With WCO, it solves two problems: clean fuel + sustainable waste management



Market Analysis

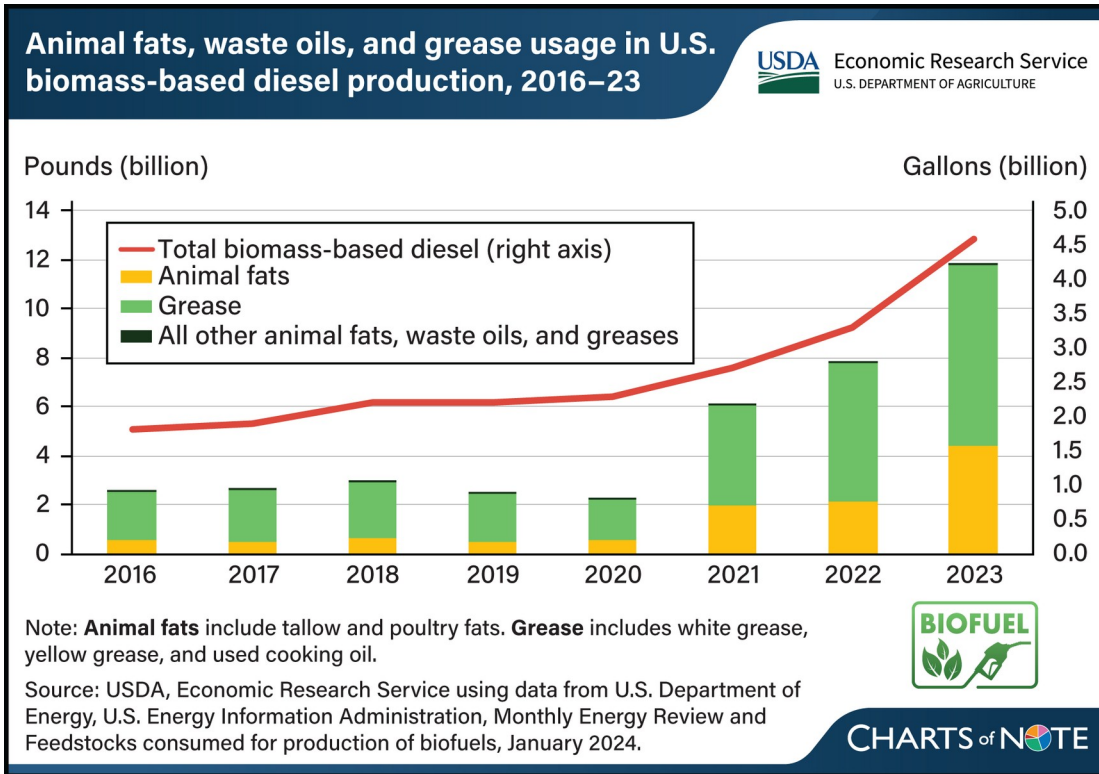
- Demand prediction
- Driven by climate commitments & transport sector decarbonization
- Major producers: US, EU, Brazil, Indonesia
- Market CAGR: ~6-10%

Biodiesel Market: Growth Rate by Region, 2024-2029



Source: Mordor Intelligence, n.d.

Market Analysis



Feedstock Availability

- Virgin oils: Soybean (US), rapeseed (EU), palm (SE Asia)
- Waste cooking oil: Abundant, 100M gal/day in US, 0.5-0.7M gal/day in EU
- Waste cooking oil = 2-3x cheaper than virgin oils
- Enables circular economy, reduces disposal problems

Market Analysis

Pricing & Competitiveness

- Feedstock = 70–95% of production cost
- Virgin oil route: ~\$3.55/gal biodiesel vs. \$2.93/gal diesel (2015)
- Current biodiesel: ~\$4 (2025)
- WCO lowers production costs → improves competitiveness

Challenges

- Higher Cost: Current biodiesel (~\$4/gal) still > diesel (~\$3/gal).
- Lower Energy Density: 8–12% less energy per gallon vs. Diesel.
- Feedstock Supply: WCO availability depends on collection & logistics.
- Processing Issues: High FFA oils need more pretreatment which means more unit operations.

Future Opportunities

- Policy incentives: (RFS2, RED II) bridge cost gap.
- Circular economy: turns waste into value.
- Tech improvements: better catalysts and recovery systems can cut costs & increase yield.
- Expanding infrastructure: improves scalability.

Process Description

Competing Processes

Soybean oil → biodiesel

- Scale

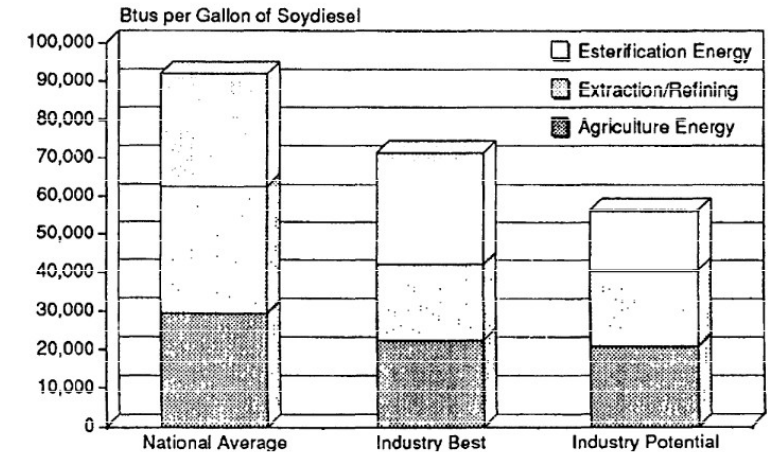
Mature national supply; dependable volumes; roughly 1.5 gal/bushel (≈50–70 gal/acre depending on yield)
- Process

Refined soy oil → base-catalyzed transesterification → B100 + glycerol; lighter pretreatment than WFO
- Cons

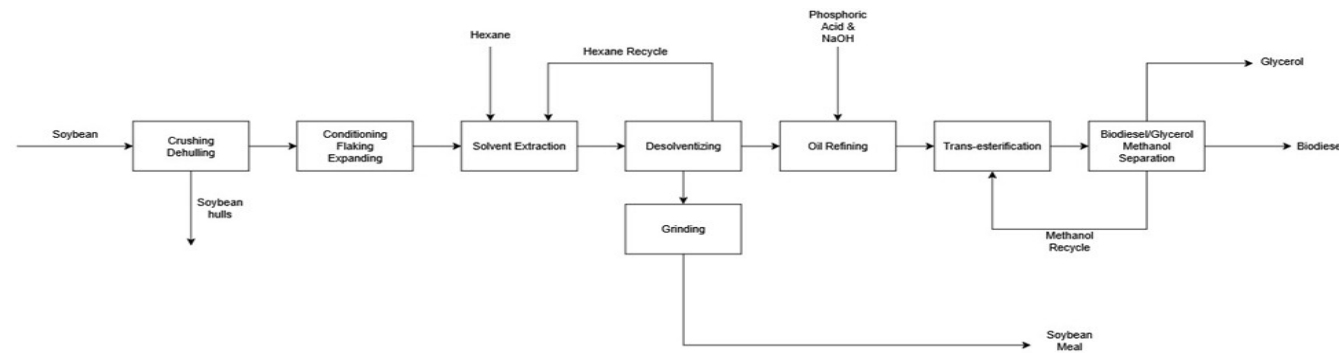
Land use and food-vs-fuel optics; profitability tied to crop and oil price swings
- Energy

Biodiesel (FAME) has ~8–12% lower energy per gallon than diesel; about the same energy as WFO biodiesel

Figure 2. Energy input Components of Soydiesel Production



NOTE: A gallon of soydiesel contains 132,902 Btus of energy. Glycerine and soy meal co-products combined contain an additional 98,239 Btus per gallon of soydiesel produced. Total energy output from the system is 231,141 Btus per gallon of soydiesel produced.



Process Description

Competing Processes

Waste tires → pyrolysis oil (TPO)

- Scale

Large, steady urban waste stream with established collection; dependable if aggregation contracts are in place

- Process

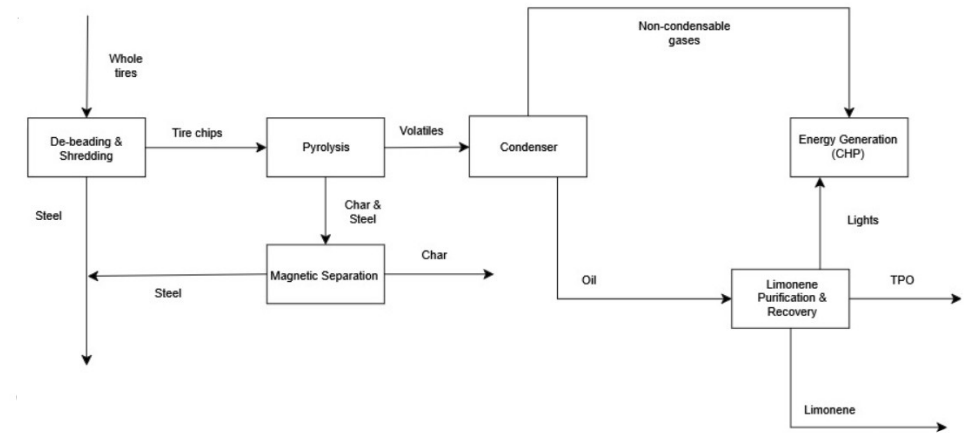
Tire shredding/steel removal → oxygen-free pyrolysis ($\approx 400\text{--}600\text{ }^{\circ}\text{C}$) → oil/char/gas; hydrotreat or blend oil; sell recovered carbon black; use process gas for heat

- Cons

Variable oil quality and sulfur; emissions permitting; need reliable offtakes for both oil and carbon black; added logistics cost

- Energy

Oil energy density near diesel and higher than biodiesel; good after cleanup/blending

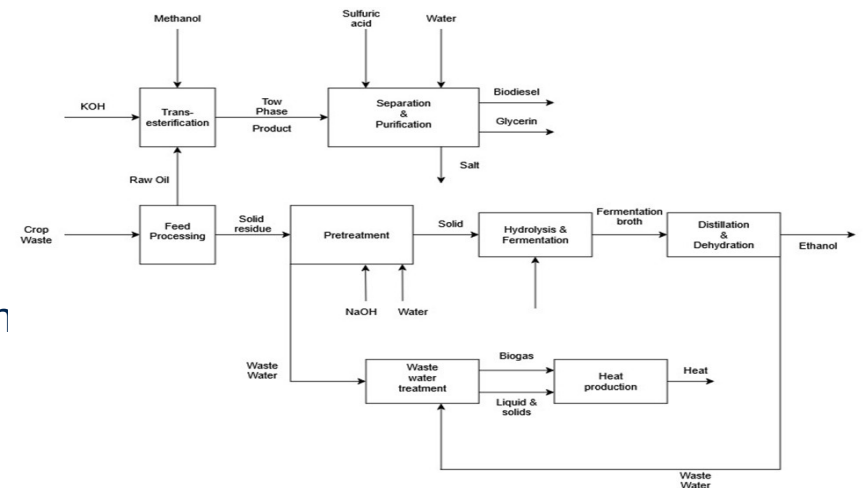
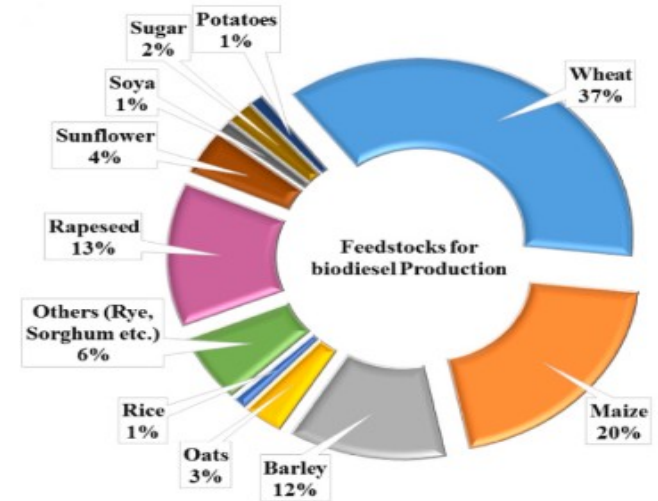


Process Description

Competing Processes

Crop residues (lignocellulosic) → biodiesel

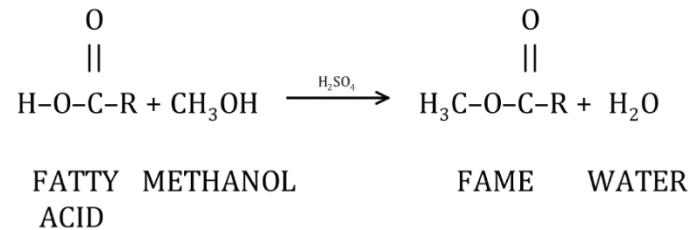
- Scale
 - Very large, widely dispersed supply; seasonal but plentiful near agricultural hubs; best with regional siting
- Process
 - Pretreatment to break biomass structure → catalytic conversion to biodiesel; more unit operations and complexity than WFO/soy
- Cons
 - Lower technology readiness today; higher CapEx and tougher chemistry; bulky feedstock logistics
- Energy
 - End fuel is biodiesel (FAME), similar energy per gallon to WFO biodiesel and ~8–12% below diesel; performance is acceptable, economics h
 - on low-cost feed and credits



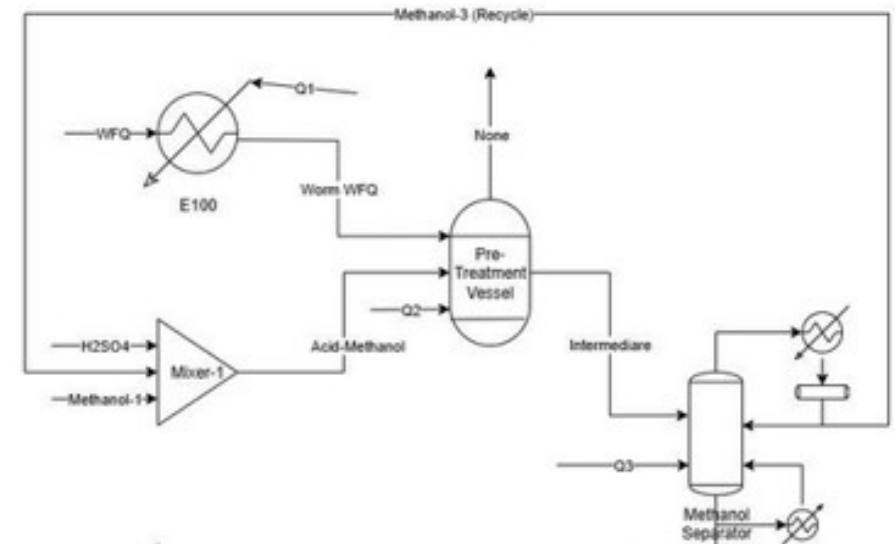
Process Description

Chosen Process: Acid Pretreatment + Base-catalyzed Transesterification

- WFO is warmed to about 55°C in heater E-100 to lower viscosity
- Oil is reacted with methanol in the Pre-treatment vessel (CSTR) to convert FFAs to FAME

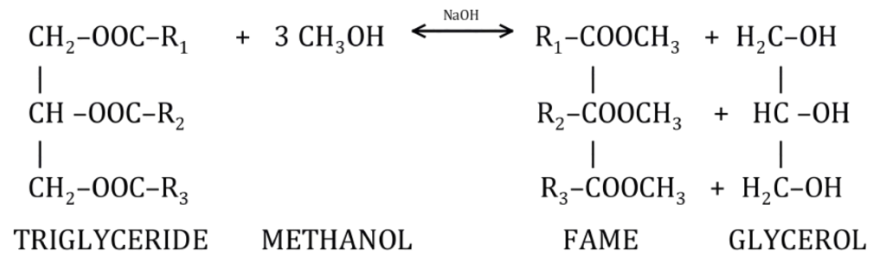


- Excess methanol is removed in Methanol Separator (Distillation Column) and ~95% gets recycled into Mixer-1

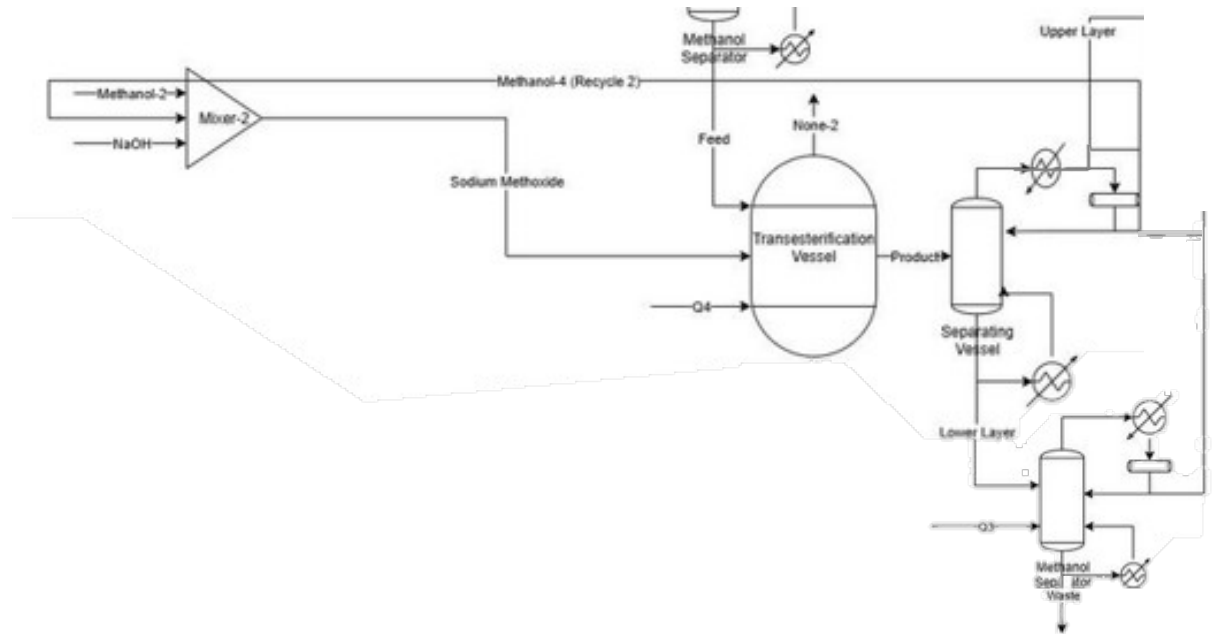


Process Description

- Fresh methanol + NaOH mixed in Mixer-2 to form sodium methoxide
- Pretreated oil reacts with sodium methoxide in Transesterification Vessel (CSTR) at 55°C.

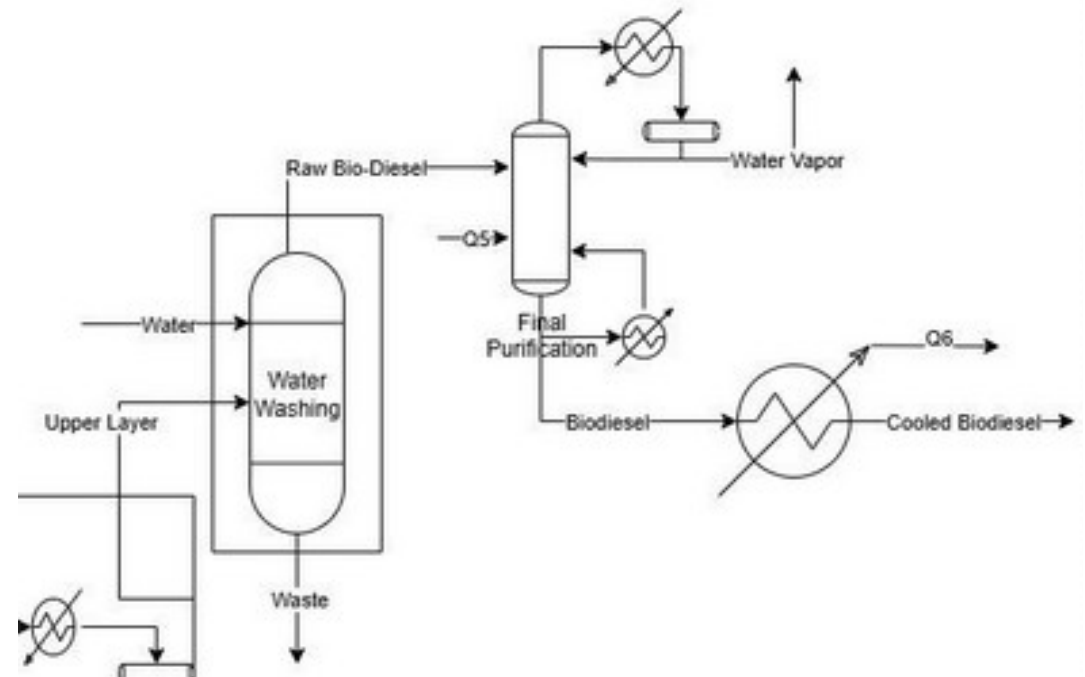


- Gravity decanter splits methanol and glycerol (lower layer) and biodiesel (upper layer)
- Lower layer gets sent to distillation column to separate methanol and recycle it to Mixer-2

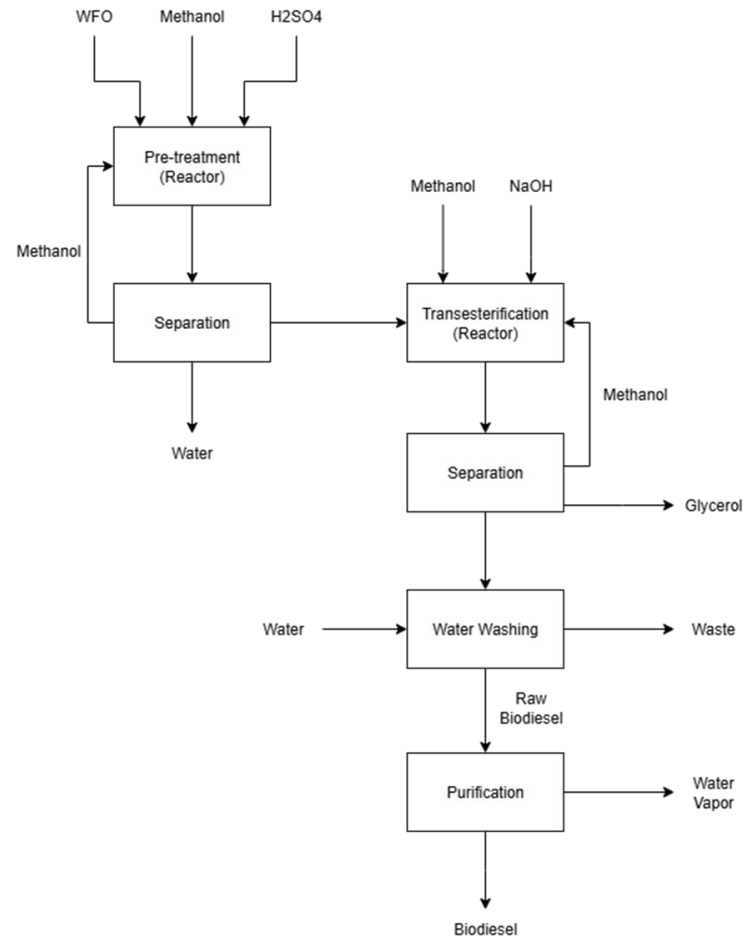


Process Description

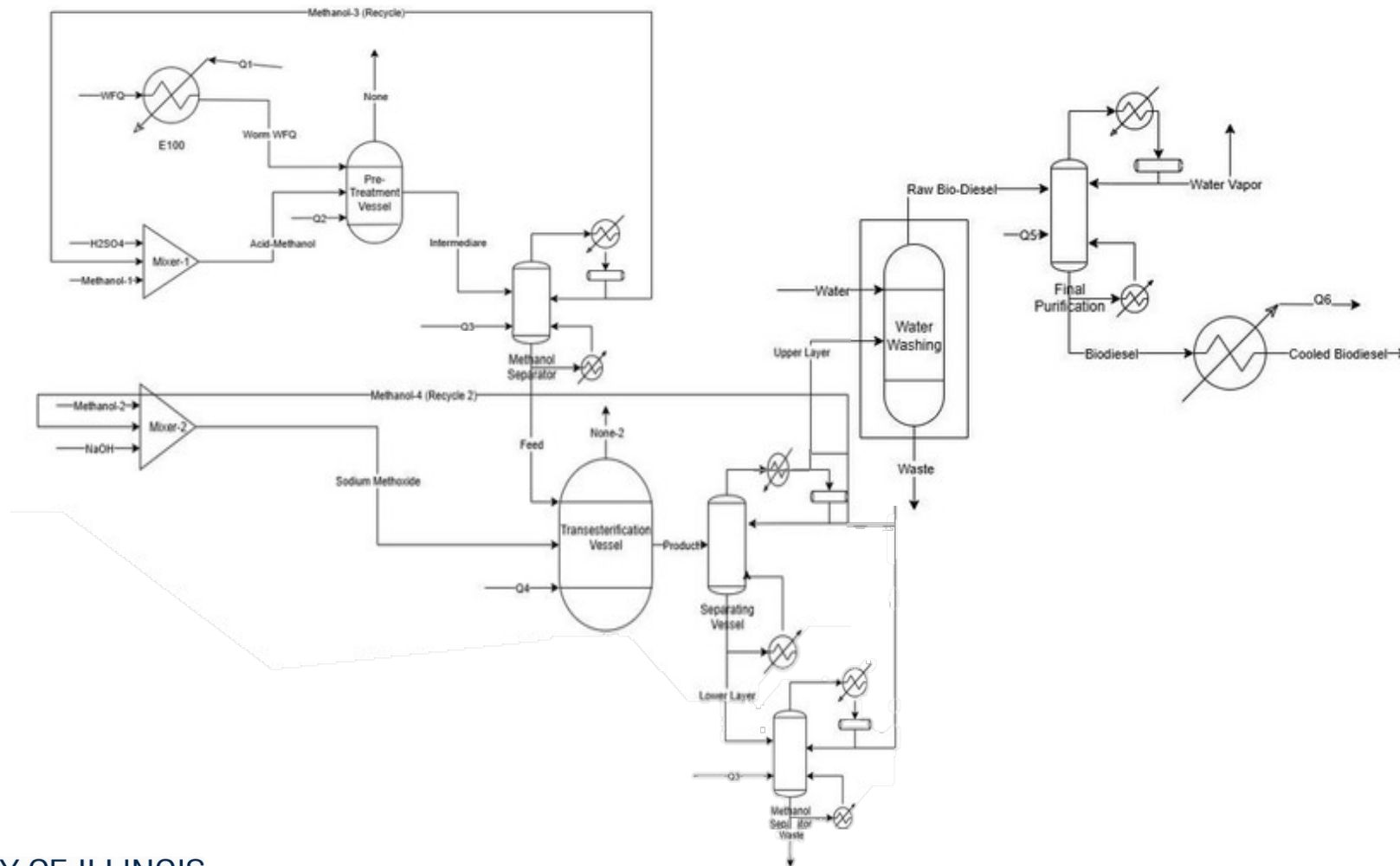
- Biodiesel (Upper Layer) is then washed counter-currently to remove any excess methanol and catalysts
- Washed biodiesel is then dried in a vacuum flash to strip any remaining water
- Final product is cooled to 30-40°C



Block Flow Diagram



Preliminary Process Flow Diagram



Rough Material Balance

Stream	From → To	Component	Flow Rate (lbmol/h)	Flow Rate (lb/hr)
S1	WCO → Pretreatment	FFA	10.31	2907
S1	WCO → Pretreatment	TG	29.56	26163
S2	MeOH Feed 1 → Pretreatment	MeOH	206.17	6606
S3	Pretreatment → Water Separator	FAME	9.79	2899
S3	Pretreatment → Water Separator	H2O	9.79	176
S3	Pretreatment → Water Separator	FFA (unreacted)	0.52	145
S4	Water Separator → Waste	H2O	9.79	176
S5	Water Separator → Base Reactor	FAME (from pretreat)	9.79	2899
S5	Water Separator → Base Reactor	FFA (unreacted)	0.52	145
S5	Water Separator → Base Reactor	TG	29.56	26163
S6	MeOH Feed 2 → Base Reactor	MeOH	177.38	5683
S7	Base Reactor → Phase Separation	FAME (new)	84.25	24939
S7	Base Reactor → Phase Separation	Glycerol	28.08	2586
S7	Base Reactor → Phase Separation	TG (unreacted)	1.48	1308
S7	Base Reactor → Phase Separation	FFA (carried)	0.52	145
S8	Phase Separation → MeOH Recovery	MeOH	383.55	12289
S9	MeOH Recovery → MeOH Header	MeOH (recycle)	364.37	11674
FRESH	Fresh MeOH → MeOH Feed Header	MeOH (makeup)	113.22	3628
S10	MeOH Recovery → Loss/Purge	MeOH (loss)	19.18	614
S11	Wash/Dry → Biodiesel Product	FAME (product)	94.04	27836
S12	Phase Separation → Glycerol Processing	Glycerol	28.08	2586

- Design Basis: 100,000 t/yr biodiesel (27,836 lb/hr)
- WCO feed: 29,070 lb/hr
- Main product: Biodiesel → meets design basis
- By-products: Glycerol, water, small unreacted oil
- Methanol recycled (95%), fresh makeup required

References

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